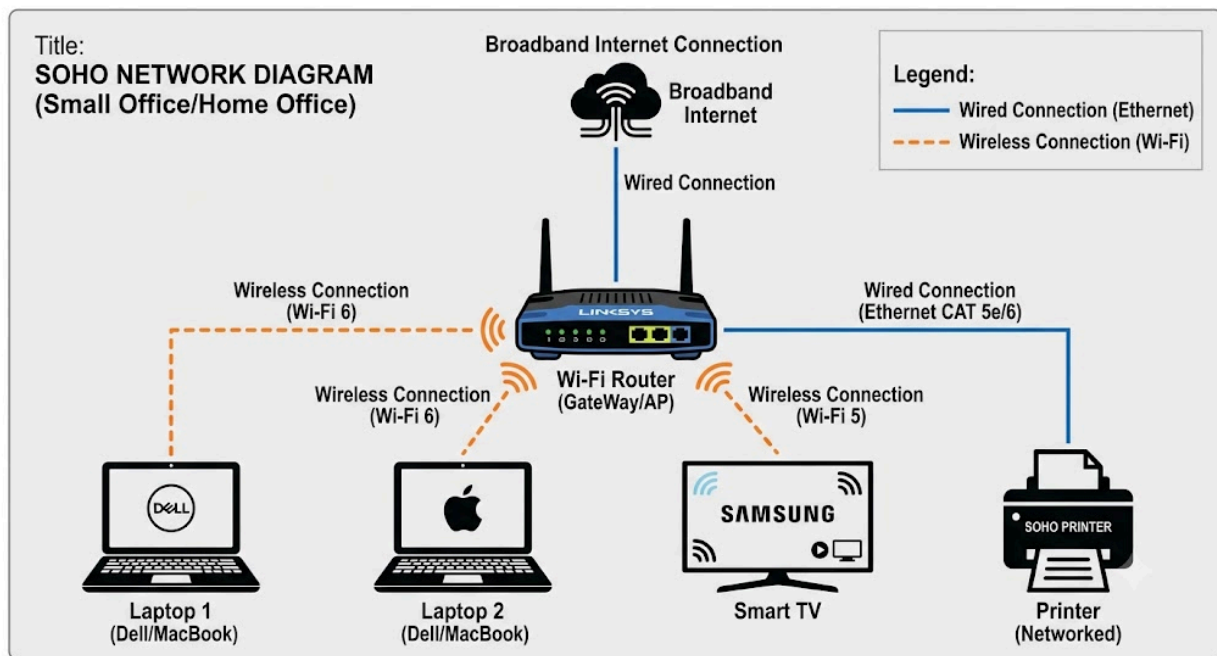


Build a SOHO Network

DAY 1

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. **Hint:** Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

ANS:

Broadband Modem / ONT: JioFiber or Airtel Xstream Fiber ONT

Wi-Fi Router: TP-Link Archer AX55 (Wi-Fi 6)

Laptops (2): Dell Inspiron / Apple MacBook Air (Connected via Wi-Fi)

Smart TV (1): Samsung 4K UHD Smart TV (Connected via Wi-Fi)

Printer (1): HP LaserJet Pro M126nw (Connected via Cat6 Ethernet cable)

Cables: 2x Cat6 RJ45 Patch Cables

Switches/Extenders: None needed (the router has built-in ports and enough coverage).

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

ANS:

Connect: Cable the modem LAN to the router WAN port, power on, and connect your laptop to a LAN port.

Access Admin Page: Open a browser and go to 192.168.0.1 (or 192.168.1.1).

Change Admin Password: Go to **System Tools > Administration** and set a strong, new password.

Set SSID & Security: Go to **Wireless Settings**, enter a custom SSID name, select **WPA2-PSK (AES)** security, and set a Wi-Fi password.

Save & Reboot: Apply settings, restart the router, and reconnect your devices to the new network.

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

ANS:

Device	Type	IP Method	IP Address	MAC Address
Router	Gateway	Static	192.168.1.1	00:1A:2B:3C:4D:01
Printer	Ethernet	Static	192.168.1.10	A4:5E:60:11:22:33
Smart TV	Wi-Fi	Static	192.168.1.20	BC:D1:D3:44:55:66
Laptop 1	Wi-Fi	DHCP	192.168.1.101	3C:A6:2F:77:88:99
Laptop 2	Wi-Fi	DHCP	192.168.1.102	FC:EC:DA:AA:BB:CC

(Subnet Mask: 255.255.255.0 | Gateway: 192.168.1.1)

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

ANS:

Disabled Wi-Fi / Airplane Mode: Toggle Wi-Fi ON or disable Airplane Mode; enable the adapter in Network Settings if needed.

Incorrect Wi-Fi Password: Select the network, click **Forget**, and reconnect with the correct password.

DHCP / IP Assignment Failure (169.254.x.x): Run `ipconfig /release` and `ipconfig /renew` in Command Prompt, or reboot the router.